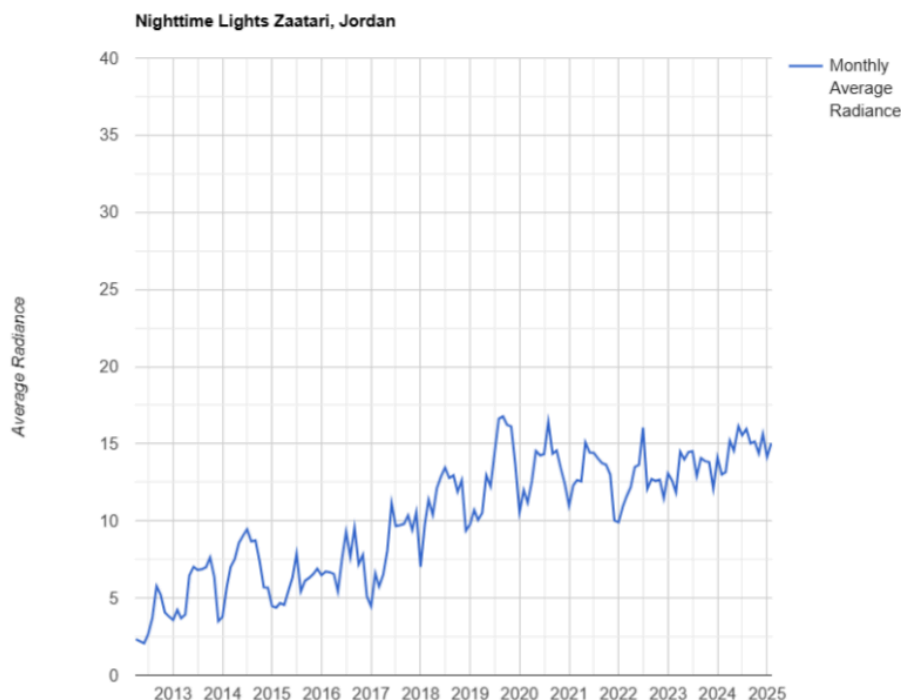


Detecting Conflict with Nighttime Lights

Nighttime lights (NTL) provide a powerful way to observe human activity, infrastructure, and environmental change from space. Because artificial lighting reflects economic development, population distribution, and energy access, NTL data can be used to study sustainability from multiple angles. Economically, increasing nighttime radiance often signals urban growth or expanding trade corridors. Socially, shifts in light patterns can reveal population displacement during crises. Environmentally, NTL can even help track changes in snowpack or surface reflectivity in colder regions.

The Syrian Civil War, which began in March 2011 during the Arab Spring, has dramatically reshaped the region's population and infrastructure. More than 500,000 people have been killed, and over 14 million Syrians have been displaced, creating one of the largest refugee crises in modern history. As cities have been damaged and depopulated, nighttime radiance has

declined sharply in many parts of Syria, while refugee settlements in neighboring countries have grown brighter. To explore these changes, I analyzed VIIRS Day/Night Band imagery for Damascus, Syria, and the Zaatari Refugee Camp in Jordan. VIIRS measures radiance in the 400–900 nm spectral range at a spatial resolution of 500 meters, making it well-suited for monitoring regional-scale changes in light. I created bounding boxes around both locations and generated monthly average radiance charts



from 2012 to 2023. The results show a clear divergence between the two sites.

Damascus begins with relatively high nighttime radiance, but its lights steadily dim over time, reflecting conflict-driven damage, depopulation, and infrastructure loss. In contrast, Zaatari starts with low radiance but grows brighter as the camp expands to accommodate increasing

numbers of refugees. The two time series often move in opposite directions- when Damascus dims, Zaatari brightens- mirroring the flow of displaced people. Spatial imagery reinforces these trends. In 2012, Damascus showed a bright, concentrated urban core, while Zaatari is barely visible. By 2023, Damascus still has a bright center, but its illuminated footprint has shrunk dramatically. Meanwhile, Zaatari has expanded into a large, well-lit settlement, with light spreading outward from the original village area. While nighttime lights are a valuable tool for detecting conflict impacts, they also come with limitations. A sudden drop in radiance may indicate population loss, but it could also result from power outages or damaged electrical infrastructure. In regions with minimal lighting to begin with, changes may be too subtle to detect. Weather, cloud cover, and sensor limitations can also affect imagery. Despite these challenges, NTL remains one of the few safe, remote methods for monitoring conflict zones where on-the-ground data collection is dangerous or impossible.

Overall, this analysis demonstrates how nighttime lights can reveal the human consequences of conflict, showing both the decline of urban centers under strain and the rapid growth of refugee settlements. The contrasting trajectories of Damascus and Zaatari offer a clear example of how remote sensing can help us understand population movement and humanitarian needs in regions affected by war.

